

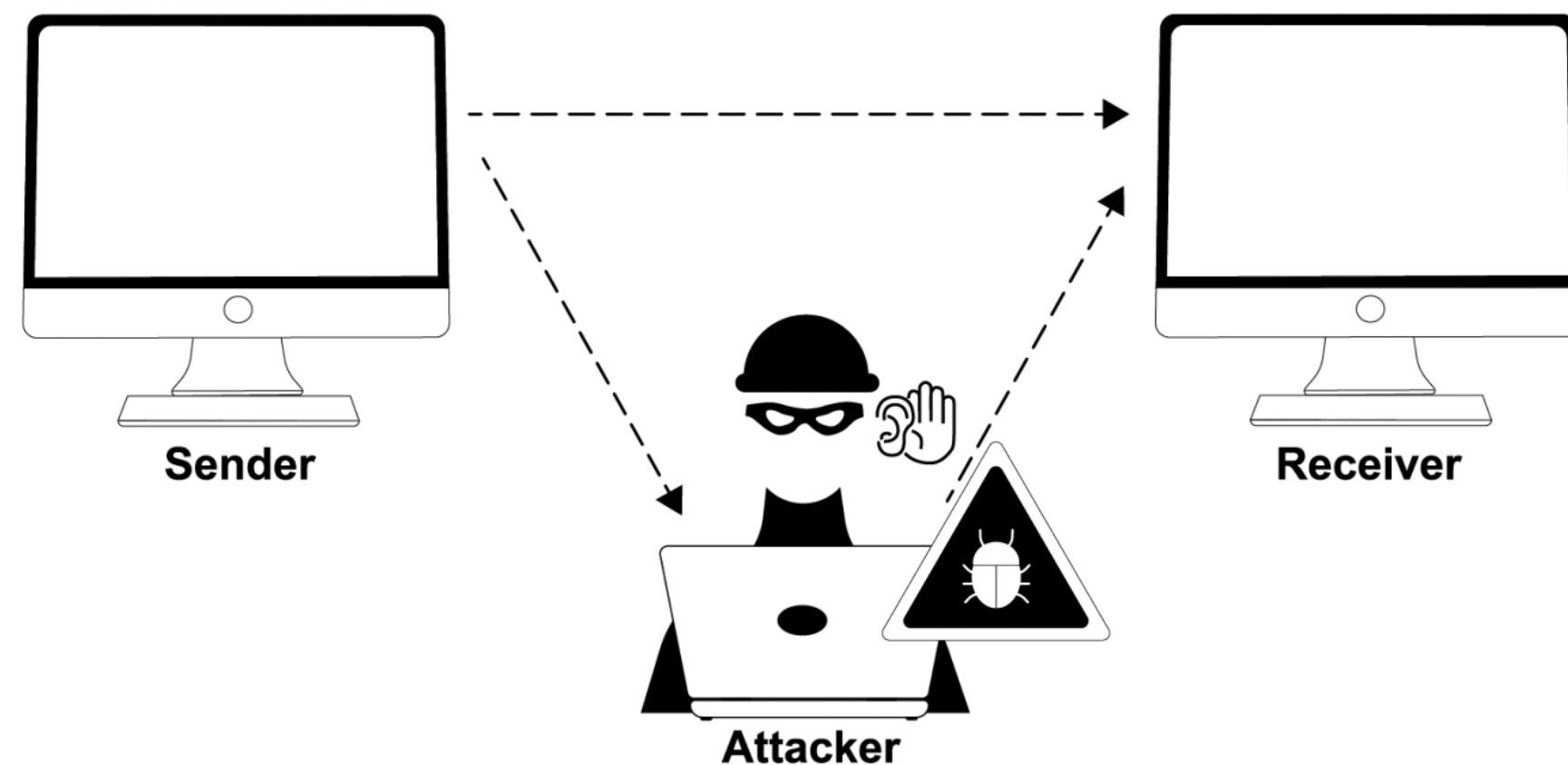
Man-in-the-middle (MITM) attacks

Learning objectives

- **Explain what MITM attacks are, how they work and how to defend against them**
- **Identify tools and techniques used by attackers**
- **Explore defences using encryption and authentication**

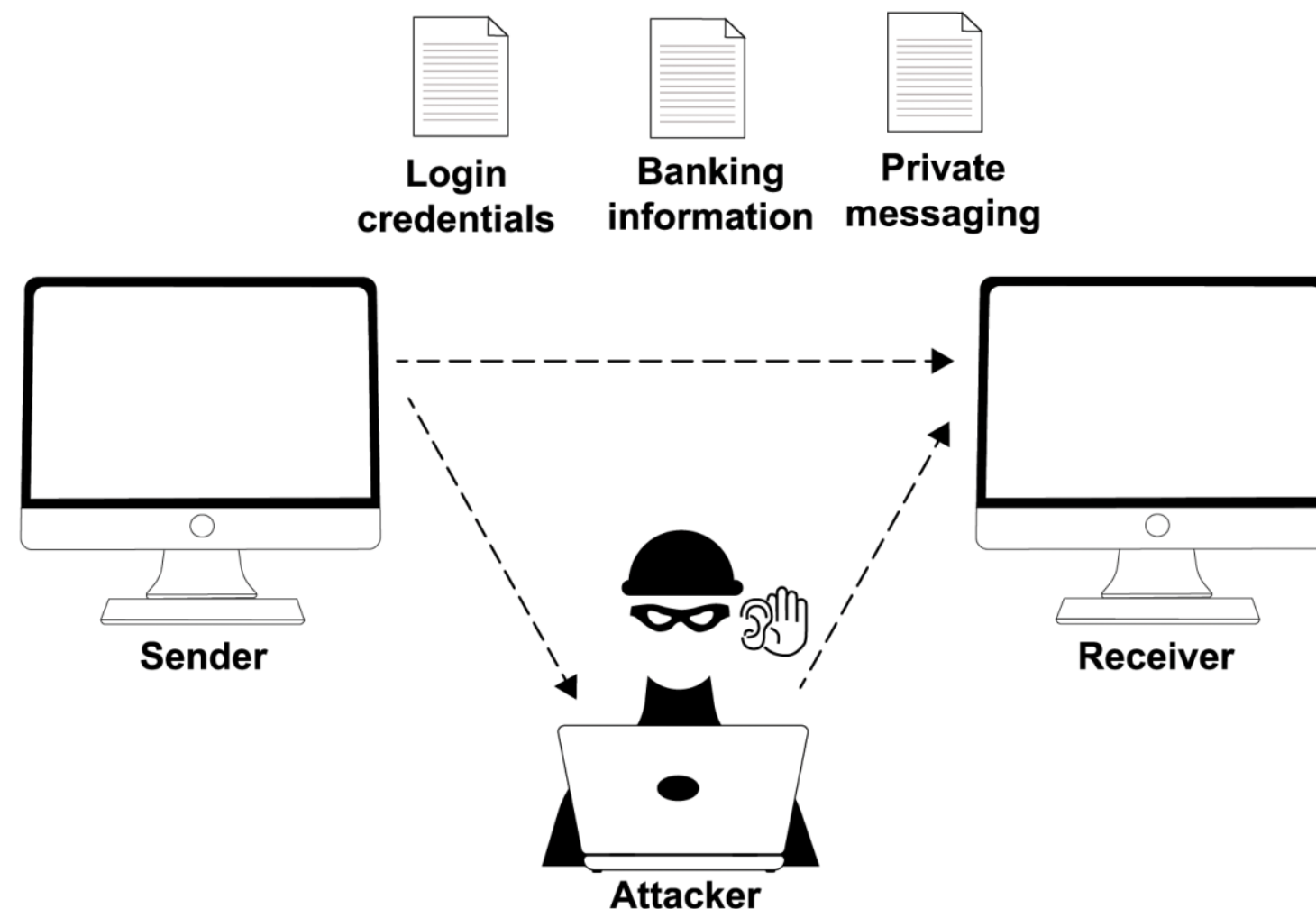
What is a MITM attack?

- **Attacker secretly intercepts communication**
- **Victim believes they are connected to a trusted party**
- **Used to eavesdrop or inject false information**



How MITM attacks work

- **Attacker places themselves between sender and receiver**
- **Captures or alters data in real time**
- **Victims often unaware of interception**



Types of MITM attacks



Wi-Fi
eavesdropping
Rogue access points



Leaked
hashes
Impersonating
trusted
addresses



DNS
spoofing
Redirecting users
to fake websites



HTTPS
stripping
Removing
encryption
from traffic

Tools used in MITM attacks

- **Ettercap: for sniffing and spoofing**
- **Wireshark: captures packets for analysis**
- **SSLstrip: removes HTTPS protection**

Real-world examples

- **Public Wi-Fi used to capture login credentials**
- **DNS spoofing is used to fake banking websites**
- **No visible signs for the victim during the attack**

How encryption helps

- **HTTPS uses TLS to encrypt browser-server traffic**
- **TLS and PGP protect email in transit**
- **Without private keys, intercepted traffic is unreadable**

Authentication and verification

- **Digital certificates validate identities**
- **Browsers warn about untrusted or expired certificates**
- **Multi-factor authentication adds an extra layer of security**

Secure network practices

- **Use VPNs on public networks**
- **Keep systems updated with security patches**
- **Educate users to avoid phishing and fake hotspots**

Summary

- **MITM attacks intercept or manipulate data silently**
- **Encryption, authentication, and awareness are key defences**

Next

- **Explore side channel attacks in modern cryptography**